LESSON 17

ESP32 SUPER STARTER KIT · 2-HOUR SESSION

Secret Agent Cards

Use RFID to read a card without touching it, and build an access-control door.

Name: _____ Date: _____

TODAY YOU WILL LEARN

- How **RFID** works (radio + a tiny chip with no battery)
- How to read a card's unique ID, called its **UID**
- How to compare **Strings** and search an **array** of allowed cards
- How to build an access system with lights, sound and (bonus) a servo lock

SESSION PLAN

0:00	How does a bus pass work? RFID explained
0:15	Wire up the RC522 reader
0:35	Sketch 1: **L17_ReadCard** — find your card's UID
0:55	Short break
1:00	Add the LEDs and buzzer
1:10	Sketch 2: **L17_Access** — allowed or denied?
1:35	Challenges (servo door!)
1:55	Quiz and tidy up

PARTS TO COLLECT

- ☐ ESP32 board + USB-C cable, 2 breadboards
- ☐ RC522 RFID module + white card + blue key tag
- ☐ 1 red LED, 1 green LED
- ☐ 2 resistors, 220 Ω (red-red-brown)
- ☐ Passive buzzer
- ☐ 7 female-to-male Dupont wires, jumper wires
- ☐ Extension: SG90 servo

1. What is RFID?

RFID stands for **R**adio **F**requency **ID**entification. It's inside bus passes, library books, hotel key cards and pet microchips.

The reader has a coil of wire (an **antenna**) that sends out a radio wave. The card also has a coil and a tiny chip, but **no battery**. When the card comes close, the radio wave makes enough electricity in its coil to wake the chip up. The chip then whispers back its number.

> **Big idea: every card has a UID**
>
> Each card has a **UID** (Unique IDentifier), a number set at the factory. Your card's UID is different from your neighbour's. Our program will decide who is allowed in by checking that number.

The UID is written in **hexadecimal** ("hex"), which counts using 16 symbols: `0 1 2 3 4 5 6 7 8 9 A B C D E F`. So a UID looks like `A3 4F 12 9B`.

2. Wire up the reader

> **3.3 V only!**
>
> The RC522 is a **3.3 V** part. Connect its power pin to **3V3**, never to 5V or VIN. 5 V can destroy it.

The RC522 talks to the ESP32 using **SPI**, a fast way of sending data over 4 wires. The ESP32 has special SPI pins, so use exactly these. Use female-to-male Dupont wires from the module's pins to the breadboard.

✓	RC522 pin	ESP32 pin
☐	**3.3V**	**3V3**
☐	**GND**	**GND**
☐	**RST**	**4**
☐	**SDA** (also called SS)	**5**
☐	**SCK**	**18**
☐	**MISO**	**19**
☐	**MOSI**	**23**
☐	**IRQ**	*not connected*

> **Install the library**
>
> **Tools → Manage Libraries…**, search **MFRC522**, and install **MFRC522 by GithubCommunity**.

3. Sketch 1: Read your card

Type this in and save it as **L17_ReadCard**. Upload it, open the **Serial Monitor** (115200 baud) and hold the card flat on the reader.

Type this in → save it as **L17_ReadCard** — 37 lines

```
// Lesson 17: Read an RFID card
// RC522 reader on SPI: SCK 18, MISO 19, MOSI 23, SDA 5, RST 4.

#include <SPI.h>
#include <MFRC522.h>

const int SS_PIN = 5;
const int RST_PIN = 4;

MFRC522 rfid(SS_PIN, RST_PIN);

String readUid() {
  String uid = "";
  for (int i = 0; i < rfid.uid.size; i++) {
    if (i > 0) uid += " ";
```

```
    if (rfid.uid.uidByte[i] < 16) uid += "0";
    uid += String(rfid.uid.uidByte[i], HEX);
  }
  uid.toUpperCase();
  return uid;
}

void setup() {
  Serial.begin(115200);
  SPI.begin();
  rfid.PCD_Init();
  Serial.println("Hold a card or tag near the reader...");
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent()) return;
  if (!rfid.PICC_ReadCardSerial()) return;

  Serial.print("Card UID: ");
  Serial.println(readUid());
  rfid.PICC_HaltA();
}
```

Write down the UIDs of your card and your key tag. You'll need them next!

Card	UID
White card	
Blue key tag	

How it works

Code	What it means
`MFRC522 rfid(SS_PIN, RST_PIN);`	Make a reader object called `rfid`
`SPI.begin();` `rfid.PCD_Init();`	Start SPI, then start the reader
`if (!rfid.PICC_IsNewCardPresent()) return;`	No card? Then `return`: stop this loop and start again. `!` means **not**
`rfid.uid.uidByte[i]`	Byte number `i` of the UID (an array of numbers)
`String(…, HEX)`	Turn a number into hex text, like `9b`
`uid += " ";`	Stick more text onto the end of the String
`uid.toUpperCase();`	`9b` becomes `9B`
`rfid.PICC_HaltA();`	Tell the card "thanks, go to sleep" so it's only read once

Try it

1. How far away can you hold the card and still read it?
2. Does it work through your hand? Through a book? Through a metal tin lid?
3. Try a bank card or a bus pass (only reading, it's safe). Do they have UIDs too?

4. Add lights and sound

Keep the reader connected and add:

✓	From	To
☐	ESP32 pin **25**	220 Ω resistor → **long leg** of the **red** LED
☐	ESP32 pin **27**	220 Ω resistor → **long leg** of the **green** LED
☐	both LED **short legs**	**–** rail
☐	ESP32 pin **33**	passive buzzer **+**
☐	buzzer **–**	**–** rail

5. Sketch 2: Access control

Save your first sketch as **L17_Access** and change it to this. **Replace the two UIDs** on the `allowed` line with the ones you wrote down. Keep the quotes and the spaces exactly.

Type this in → save it as **L17_Access** 64 lines

```
// Lesson 17: Access control
// Allowed cards light green and beep once, others red and buzz.

#include <SPI.h>
#include <MFRC522.h>

const int SS_PIN = 5;
const int RST_PIN = 4;
const int RED_PIN = 25;
const int GREEN_PIN = 27;
const int BUZZER_PIN = 33;

// Put your own card UIDs here (copy them from L17_ReadCard)
String allowed[] = {"A3 4F 12 9B", "04 7E 2C 1A"};
const int NUM_ALLOWED = 2;

MFRC522 rfid(SS_PIN, RST_PIN);

String readUid() {
  String uid = "";
  for (int i = 0; i < rfid.uid.size; i++) {
    if (i > 0) uid += " ";
    if (rfid.uid.uidByte[i] < 16) uid += "0";
```

```
    uid += String(rfid.uid.uidByte[i], HEX);
  }
  uid.toUpperCase();
  return uid;
}

bool isAllowed(String uid) {
  for (int i = 0; i < NUM_ALLOWED; i++) {
    if (uid == allowed[i]) return true;
  }
  return false;
}

void setup() {
  Serial.begin(115200);
  pinMode(RED_PIN, OUTPUT);
  pinMode(GREEN_PIN, OUTPUT);
  SPI.begin();
  rfid.PCD_Init();
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent()) return;
  if (!rfid.PICC_ReadCardSerial()) return;

  String uid = readUid();
  Serial.print(uid);
  if (isAllowed(uid)) {
    Serial.println("  -> Welcome!");
    digitalWrite(GREEN_PIN, HIGH);
    tone(BUZZER_PIN, 1500, 150);
  } else {
    Serial.println("  -> Access denied");
    digitalWrite(RED_PIN, HIGH);
    tone(BUZZER_PIN, 200, 800);
  }
  delay(2000);
  digitalWrite(GREEN_PIN, LOW);
  digitalWrite(RED_PIN, LOW);
  rfid.PICC_HaltA();
}
```

How it works

Code	What it means
`String allowed[] = {"…", "…"};`	An **array of Strings**: the list of cards that may enter
`bool isAllowed(String uid)`	A function that answers **true** or **false**
`if (uid == allowed[i]) return true;`	`==` compares two Strings. Found it? Answer true straight away
`return false;`	Checked the whole list and no match: answer false
`tone(BUZZER_PIN, 1500, 150);`	High beep for 150 ms. A low 200 Hz buzz sounds like "nope!"

Every card says denied?

Print the UID and compare it letter by letter with your list. Look for `0` vs `O`, missing spaces, or lowercase letters.

6. Challenges

Challenge 1: Add a friend

Add a third card (borrow a friend's) to the list. Remember to change `NUM_ALLOWED` too!

Challenge 2: Say hello

Make a second array `names[]` with a name for each card, and print `Welcome, Sam!` with the right name.

Challenge 3: Servo door

Add a servo as a door lock: it turns to 90° for 3 seconds when an allowed card is shown. Pin 18 is busy (SPI clock), so put the servo's signal wire on pin **13**, red on **5V**, brown on **GND**.

Bonus: Intruder alarm

Count wrong cards in a row. After 3 wrong cards, flash the red LED and sound a siren.

7. Quiz

Check what you learned

1. Why doesn't an RFID card need a battery?

2. Why must the RC522 be connected to 3V3 and not 5V?

3. What does `return` do inside `loop()`?

4. Is a UID a good password? Why or why not? *(Hint: could someone copy it?)*

Tidy up: unplug the USB cable. Put the card and tag back in the kit. You can leave the reader wired up.